

Bernardo Lanza

Ph.D. | Robotics Engineer



About me

Robotics & sensing engineer (Ph.D. mechatronics) specialized in vision-based systems for precision agriculture: tractor-mounted sensing, in-field validation, 3D orchard reconstruction. Also active in industrial digital twins. Outside work: history, DIY IoT, volunteering.

Areas of specialization

Metrology • Statistics • Optical sensors • Embedded Linux • Computer Science • Dynamics • Physics

Soft skill

Cross-functional collaboration • Rapid prototyping • Scientific communication • Field problem-solving

Interests

Blockchain • Electronics • Arduino • Biology • History • Volunteering

LANGUAGES

Italian | Native speaker
English | Professional working proficiency

 bernardolanza
 bernardolanza93



Full Portfolio

PROFESSIONAL EXPERIENCE

- 2025–26 **Pink Peak**
SOFTWARE DEVELOPER ENGINEER · Remote 
Event-driven backend (pub/sub broker, time-series). Predictive maintenance: thermodynamic digital twin for cold chain, sensor-driven feature engineering. 
- 2022–24 **Ph.D.**
PH.D. RESEARCHER · Mechanical and Thermal Measurement Lab 
Designed and validated tractor-mounted vision systems for vineyard monitoring: optical-flow depth estimation ($u < 0.04$ m, 20 test configs) and real-time bud detection (YOLOv8, F1 = 0.79) 
- 2024 **Ph.D. abroad: NextGenEU project**
VISITING PH.D. RESEARCHER · ETSEA, University of Lleida Spain 
Multi-sensor 3D orchard reconstruction (LiDAR + RGB-D + GNSS/IMU). Custom SLAM validated against 6 metrological reference targets 
- 2023 **Prospecto**
EMBEDDED DEVELOPER, PH.D STAGE · Brescia 
Developed embedded sensing pipeline (MCU + sensors) for real-time agricultural monitoring 
- 2021–22 **Research Fellowship**
RESEARCH SCIENTIST · University of Brescia 
Jetson Nano vision pipelines: CNN for crop/weed segmentation and pose estimation, both field-validated. 

CERTIFICATES

- 2025 **Computer vision for agriculture**
PH.D. · University of Brescia
- 2023 **IEEE Best poster awards**
AWARD · Metrology for Agriculture and Forestry
- 2023 **"Sensor Fusion: Naïve Bayes to Kalman Filters"**
GUEST LECTURE · University of Brescia
- 2021 **DeepLearn 2021 Summer school**
IRTD · International School on Deep Learning
- 2020 **Mechatronic Engineering**
M.S. · University of Trento
- 2017 **Industrial Engineering**
B.S. · University of Trento

PROJECTS

- 2023 **Gesture recognition for Healthcare 4.0**
RESEARCH SCIENTIST · University of Brescia 
Vision-based smart sensor for real-time handwashing detection. Multi-stage ML (Random Forest + ANN) with in-field validation.
- 2022 **Embedded Pose Estimation System for Gym Gesture Recognition**
VISION SYSTEMS DEVELOPER · University of Brescia 
Jetson Nano + dual fisheye, validated against motion capture ground truth.

PUBLICATIONS

- 2024 *A Stride Toward Wine Yield Estimation from Images: Metrological Validation of Grape Berry Number, Radius, and Volume Estimation*, MDPI Sensors
- 2023 *First Step Towards Embedded Vision System for Pruning Wood Estimation*, IEEE Metrology for Agriculture and Forestry
- 2023 *Gesture recognition for Healthcare 4.0: a machine learning approach to reduce clinical infection risks*, IEEE Xplore.

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